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Beverage container packaging

Abstract

A beverage container packaging assembly is disclosed. The beverage container packaging assembly includes a bottom tray, center support, and a top tray. The bottom tray includes a plurality of cup-shaped elements each configured to accommodate a bottom portion of a beverage container. Each cup-shaped element includes at least a first deformable element comprising one or more wide portions and one or more thin portions. The center support is disposed between the bottom tray and the top tray. The top tray includes bottle neck accommodating spaces each bounded by a set of vertical posts.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. patent application Ser. No. 17/397,619, filed Aug. 9, 2021, which is a continuation of U.S. patent application Ser. No. 16/884,922, filed on May 27, 2020, which issued as U.S. Pat. No. 11,104,471 on Aug. 11, 2021, which is a continuation of U.S. patent application Ser. No. 16/180,840, filed Nov. 5, 2018, and which issued as U.S. Pat. No. 10,696,441 on Jun. 30, 2020, which is a continuation of U.S. patent application Ser. No. 15/671,348 filed Aug. 8, 2017, which issued as U.S. Pat. No. 10,124,924 on Nov. 13, 2018, which claims priority to U.S. Provisional Application No. 62/372,129, filed on Aug. 8, 2016, the entire contents of the disclosures of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present invention relates to packaging materials, and particularly to packaging for beverage containers, such as wine bottles, beer bottles, spirits bottles, and the like.

BACKGROUND

(2) Historically, packaging materials for shipment of beverage containers (e.g., wine bottles, beer bottles, spirits bottles, etc.) have included bubble wrap, Styrofoam, popcorn, and other traditional packaging materials. For example, multiple bottles could be wrapped in bubble wrap, positioned in Styrofoam, and/or otherwise secured and placed into a box for transit. More recently, molded paper pulp trays have been used to secure multiple bottles during transit. Typically, a bottom tray accommodates the bottom of a bottle, a top tray accommodates the top of the bottle, and cardboard may be installed between the bottles. Many existing bottom tray designs include ring-shaped crushable elements to absorb impact during shipping. The ring-shaped crushable elements may be ineffective in reducing and/or preventing damage to the bottle when the package is subjected to a large impact and/or multiple large impacts. An improved beverage container packaging would be useful.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The foregoing and other features and advantages of the invention will be apparent from the following, more particular description of various exemplary embodiments, as illustrated in the accompanying drawings wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. The first digits in the reference number indicate the drawing in which an element first appears.

- (2) FIG. 1 depicts a beverage container packaging assembly according to various embodiments.
- (3) FIG. 2 depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments.
- (4) FIG. 3 depicts a bottom tray of a beverage container packaging assembly according to some embodiments.
- (5) FIG. 4 depicts a first view of a bottom tray according to various embodiments.
- (6) FIG. 5 depicts a second view of a bottom tray according to various embodiments.
- (7) FIG. 6 depicts a center support of a beverage container packaging assembly according to various embodiments.
- (8) FIG. 7 depicts a bottom side of a center support according to various embodiments.
- (9) FIG. 8 depicts a top tray of a beverage container packaging assembly according to some embodiments.
- (10) FIG. 9 depicts a first view of a top tray according to some embodiments.
- (11) FIG. 10 depicts a second view of a top tray according to various embodiments.
- (12) FIG. 11 depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments.
- (13) FIG. 12 depicts a bottom tray according to various embodiments.
- (14) FIG. 13 depicts a center support according to various embodiments.
- (15) FIG. 14 depicts a top tray according to various embodiments.
- (16) FIG. 15 depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments.
- (17) FIG. 16 depicts a center support according to various embodiments.
- (18) FIG. 17 depicts a bottom view of a center support according to various embodiments.
- (19) FIG. 18 depicts a beverage container packaging assembly according to various embodiments.
- (20) FIG. 19 depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments.
- (21) FIG. 20 depicts a bottom tray of a beverage container packaging assembly according to some embodiments.
- (22) FIG. 21 depicts cup-shaped elements of a bottom tray according to various embodiments.
- (23) FIG. 22 depicts a bottom tray of a beverage container packaging assembly according to various embodiments.
- (24) FIG. 23 depicts a first cross-section view of a cup-shaped element of a bottom tray according to various embodiments.
- (25) FIG. 24 depicts a second cross-section view of a cup-shaped element of a bottom tray according to various embodiments.
- (26) FIG. 25 depicts a top tray of a beverage container packaging assembly according to some embodiments.
- (27) FIG. 26 depicts a top tray of a beverage container packaging assembly according to some embodiments.
- (28) FIG. 27 depicts a bottom tray and center support of a beverage container packaging assembly according to some embodiments.
- (29) FIG. 28 depicts a beverage container packaging assembly including a lower tray and center support according to some embodiments.

DETAILED DESCRIPTION

(30) Exemplary embodiments are discussed in detail below. While specific exemplary embodiments are discussed, it should be understood that this is done for illustration purposes only. In describing and illustrating the exemplary embodiments, specific terminology is employed for the sake of clarity. However, the embodiments are not intended to be limited to the specific terminology so selected. A person skilled in the relevant art will recognize that other components and configurations may be used without parting from the spirit and scope of the embodiments. It is

to be understood that each specific element includes all technical equivalents that operate in a similar manner to accomplish a similar purpose. The examples and embodiments described herein are non-limiting examples.

(31) All publications and references cited herein are hereby incorporated by reference in their entirety.

(32) As used herein, the term “a” refers to one or more. The terms “including,” “for example,” “such as,” “e.g.,” “may be” and the like, are meant to include, but are not be limited to, the listed examples.

(33) Beverage container packaging is disclosed. The beverage container packaging disclosed herein in various embodiments may include a bottom element (bottom tray, bottom insert), a top element (top tray, top insert), and/or a center support (e.g., center support element). These elements may retain multiple beverage containers, such as wine bottles, beer bottles, spirits bottles, and the like, for shipment in a container, such as a cardboard box, crate, or other container. The beverage containers may vary in size including, for example, 375 ml, 500 ml, 750 ml, or any other size. The term beverage container as used herein may also refer to any container that encloses a fluid, whether or not the fluid is a beverage. Beverage containers may be placed bottom down in the bottom tray. The center support rests on top of the bottom tray and separates the multiple bottles. The center support separates the bottom tray and top tray, thereby providing stacking support. The top tray rests on the center support. And the top element accommodates the top (neck) of the beverage containers. The bottom tray, center support, top tray, and beverage containers are placed in a container, such as a cardboard box, crate, etc., for transit.

(34) In various embodiments, the bottom tray comprises molded paper pulp fabricated from, for example, recycled paper products. The bottom tray may include multiple cup-shaped elements (e.g., cup-shaped cavities) each contoured to encapsulate the bottom of a beverage container, such as a wine bottle, beer bottle, spirits bottles, or the like. In certain cases, the cylindrical cup-shaped element includes an hourglass-shaped deformable element and/or a cross shaped deformable element protruding from the bottom surface of the cup, deformable protrusions on the walls of the cup, and/or other features. These features retain the base of the beverage container in a stationary position during shipment by contacting the base of the container in multiple locations. These features of the cup-shaped element, particularly the hourglass shaped and/or cross-shaped deformable element absorb energy when the container is subjected to impact forces (e.g., when dropped, roughly handled, etc.).

(35) In some embodiments, the top tray comprises molded paper pulp. In certain cases, the top tray may include multiple rectangular cup-shaped elements each contoured to encapsulate the top of a beverage container, such as the neck of a wine bottle, beer bottle, spirits bottle, or other container. The rectangular cup-shaped element may include a cylindrical depression, vertical protrusions on the walls of the cup, and/or other features. These features retain the neck and/or upper portion of the beverage container in a stationary position during shipment by contacting the neck of the container in multiple locations.

(36) In various embodiments, the center support may comprise one or more sheets of cardboard, such as corrugated cardboard. In some cases, the center support may comprise molded paper pulp that is molded to encapsulate beverage containers.

(37) The beverage container packaging assembly disclosed herein in various embodiments provides improved impact energy absorption characteristics in relation to existing packaging solutions. The beverage container packaging disclosed herein may also be cheaper to produce than existing packaging solutions.

(38) FIG. 1 depicts a beverage container packaging assembly according to various embodiments. In the example shown, a beverage container packaging assembly **100** may include a bottom tray **110** (e.g., bottom element), a top tray **120** (e.g., a top element), a center support **130** (e.g., center support element, partition element), and/or other components. The beverage container packaging

assembly **100** is configured to package a plurality of beverage containers **140** for transit. In the example shown, the beverage container packaging assembly **100** is configured to package **12** bottles. The base of the beverage container **140** sits in a cylindrical cup-shaped elements of the bottom tray **110**. The center support **130** separates the bottom tray **110** and top tray **120**. The top tray **120** rests on the center support **130**. The top of the beverage container **140** is encapsulated in a bottle accommodating space of the top tray **120**. The center support **130** prevents the sides of the beverage containers **140** from contacting during shipping and provides spacing between the bottom tray **110** and top tray **120**. In certain cases, the center support **130** may not be included in the beverage container packaging assembly **100**.

(39) FIG. 2 depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments. In the example shown, a beverage container **140** (e.g., wine bottle) is seated in a cup-shaped element **112** of a bottom tray **110**. The cup-shaped element **112** is shaped to retain the beverage container **140** in a vertical configuration during transit. In the example shown, a center support **130** is fabricated from molded paper pulp and/or similar materials. The center support **130** includes exterior support cavities **132**, interior support cavities **134**, and/or other elements.

(40) In some embodiments, the center support **130** does not extend to an outer edge **150** of the bottom tray **110** and/or an outer edge of the bottle container packaging assembly (e.g., bottle container packaging assembly **100** of FIG. 1). In this case, the center support **130** is located on an interior portion of the beverage container packaging assembly and does not contact the box. A center support **130** that does not next extend to an outer edge **150** of the bottom tray **110** may require less material than other large center support designs.

(41) As discussed below, other types of center supports, such as corrugated cardboard center supports, may be used. In certain cases, a type of center support may be selected based on shipping requirements, load absorption parameters, customer preferences, and/or any other parameters.

(42) FIG. 3 depicts a bottom tray of a beverage container packaging assembly according to some embodiments. As shown, a bottom tray **110** includes a plurality of cup-shaped elements **112**. The number of cup-shaped elements **112** corresponds to a number of beverage containers the bottom tray **110** is configured to accommodate. In the example shown, the bottom tray **110** includes twelve cup-shaped elements **112**—three (3) rows of four (4) cup-shaped elements **112**. The bottom tray **110** is thus configured to accommodate 12 beverage containers. FIG. 3 depicts one example configuration of cup-shaped elements. The present disclosure, however, is in no way limited to the depicted configuration or number of cup-shaped elements **112**.

(43) In certain cases, the bottom tray **110** includes posts **114** configured to support the center support (not shown). Each of the posts **114** (six in the example shown) may include a plurality of platforms **116** (e.g., deformable post elements, protrusions, etc.) that form flat surfaces to contact the center support. The platforms **116** may be arranged in sets to resemble a cross shape. The platforms **116** may also be deformable and/or flexible to absorb load applied to the bottom tray **110**, thereby reducing any load applied to the beverage containers **140**.

(44) FIG. 4 depicts a first view of a bottom tray according to various embodiments. In the example shown, a bottom tray **400** includes a plurality of cup-shaped elements **410** (e.g., cup shaped element **112** of FIGS. 1-3). In certain cases, each of the cup-shaped elements **410** is cylindrical and/or substantially cylindrical. The cup-shaped element **410** may include a bottom portion and multiple side walls forming a cup shape. In certain cases, the side walls may not be vertical but may include draft, such a seven degree draft, to enable the bottom tray **400** to be easily removed from a mold during fabrication.

(45) Each of multiple cup-shaped elements **410** may include one or more of deformable elements **420** on the base (bottom) of the cup-shaped elements **410**, first deformable elements **430** on the cup-shaped elements **410**, second deformable elements **440** on side(s) of the cup-shaped elements **410**, thin deformable elements **450** on sides of the cup-shaped elements **410**, and/or other

components.

(46) The deformable elements **420** (e.g., base deformable elements) on the base (bottom) of the cup-shaped elements **410** may be raised (e.g., protrusions) from the bottom surface of the cup-shaped element **410**. The deformable elements **420** on the bottom of the cup-shaped element **410** may resemble a cross, overlapping bowties, and/or other similar shapes. The deformable elements **420** are configured to absorb impact with the bottom of a beverage container, such as a load applied to a top or bottom of a beverage container when boxes are stacked, a box is dropped, and/or otherwise handled. The cross-shaped deformable element **420** may be particularly effective in absorbing impact on a bottle with a convex bottom shape, such as certain wine bottles. In certain cases, wider portions **422** of the cross-shaped deformable element **420** are configured to contact the outer edges of the bottom of a beverage container, such as a wine bottle. Thinner portions **424** of the cross-shaped deformable element **420** allow the element to deform (for example, by bending, crumpling, and/or otherwise deforming) when a load is applied to a beverage container housed in the bottom tray **400**. The cross-shaped deformable element **420** deforms to absorb load and reduce damage to a bottle, particularly when a load is applied down from the top on the bottle or when the assembly is dropped.

(47) In some embodiments, deformable elements **420** on a bottom of the cup-shaped elements **410** include holes **426**. The holes **426** may vent air as the bottom tray **400** is lowered into a box (not shown). Vent holes **426** may reduce the vacuum in the box as the bottom tray **400** is installed in a box, thereby making assembly easier.

(48) In various embodiments, the cup-shaped element **410** includes multiple vertical deformable elements **430**, **440**. The vertical deformable elements **430**, **440** are configured to contact the sides of a beverage container (not shown). In certain cases, first vertical elements **430** and second vertical elements **440** are configured to contact the outside of a bottle. The vertical deformable elements **430**, **440** may be sized, such that a bottle contacts at least a portion of each of the vertical deformable elements **430**, **440** when loaded into the cup-shaped element **410**. The bottle (not shown) and vertical deformable elements **430**, **440** may contact one another in an interference fit, such that a force is necessary to push the bottle into the cup-shaped element **410** and a force is necessary to remove the bottle from the cup-shaped element **410**. Securing the bottle in the cup-shaped element **410** in such a manner ensures that the bottle is stationary during transit, thereby reducing any potential damage. In certain cases, the vertical deformable elements **430**, **440** can include a protrusions **460** near the bottom of the cup-shaped element **410** extending toward a center of the cup-shaped element **410**. The protrusions **460** may increase the interference fit between the cup-shaped element **410** and a bottle.

(49) In certain cases, a first type of vertical deformable elements **430** are included on the walls between adjacent cup-shaped elements **410**. The first vertical elements **430** include a contoured protrusion configured to contact the outside of a bottle.

(50) In some embodiments, a second type of vertical deformable elements **440** are included on outer walls of the cups-shaped element **410** (e.g., walls of the cup-shaped elements not adjacent to any other cup-shaped element **410**). A lower portion of the second vertical deformable elements **440** may resemble the structure of the first vertical deformable elements **430**, and an upper portion may include multiple thinner protrusions **442**. In the example shown, the multiple thin protrusions **442** may resemble a fork.

(51) According to some embodiments, the cup-shaped element **410** includes multiple thin deformable elements **450**. In certain cases, thin deformable elements **450** may extend from a center of the cup-shaped element up a side of the cup-shaped element **410**. Thin deformable elements **450** may include thin protrusions spaced roughly 90 degrees apart from one another. In certain cases, the thin deformable elements **450** may be spaced roughly 45 degrees from the intersection of the cross-shaped deformable element **420**. In the example shown, there are four thin deformable elements **450** in each cup shaped element **410**. Though in other cases, a cup-shaped element **410**

may include other numbers of first thin deformable elements **450**.

(52) FIG. 5 depicts a second view of a bottom tray according to various embodiments. In the example shown, which may include an opposite side of the bottom tray from FIG. 4, a bottom tray **500** includes multiple cup-shaped elements **510**. The bottom side of the bottom tray **500** includes non-contoured pulp paper. The bottom side of the cup-shaped elements **510** may include protrusions **520** (e.g., four protrusions in the example shown) extending from the bottom of the cup-shaped element **510** up each side. The protrusions **520** may absorb impact and reduce loads applied to the beverage containers during, for example, an impact event.

(53) FIG. 6 depicts a center support of a beverage container packaging assembly according to various embodiments. In the example shown, a center support **600** includes a plurality of exterior support cavities **610** (e.g., beverage container support cavities), interior support cavities **620**, and/or other elements. The exterior support cavities **610** and interior support cavities **620** maintain the bottle upright and to separate each bottle from adjacent bottles. Exterior support cavities **610** may include a semi-circular shape that follows the contour of a wine bottle. An exterior support cavity **610** encapsulates a portion of a beverage container (e.g., a body, shoulder, or neck of a bottle). In certain cases, an exterior support cavity **610** includes walls **612** including a draft angle of seven degrees and/or another draft angle. The interior support cavities **620** may include a circular (cylindrical) enclosure. The interior support cavities **620** may fully encapsulate (surround) a neck, shoulder, body and/or other component of a beverage container. Walls **622** of the interior support cavities **620** may include a draft angle of seven degrees and/or another draft angle.

(54) In some embodiments, the center support **600** includes star shaped cavities **630** (e.g., cross shaped cavities). The star-shaped cavities **630** may form the borders of the exterior support cavities **610**, interior support cavities **620**, and/or other elements. In certain cases, the posts of a top tray (discussed below) contact the corners **632** of the star-shaped cavities **630** when assembled. In certain cases, the corners **632** of the star-shaped cavities **630** extend out away from the center far enough to provide cushion between adjacent beverage containers. For example, the corners **632** of the star-shaped cavities **630** can extend out away from the center far enough to encapsulate at least half of the circumference of a bottle.

(55) FIG. 7 depicts a bottom side of a center support according to various embodiments. In the example shown, a bottom side of center support **700** (e.g., center support **600** of FIG. 6, center support **130** of FIGS. 1-2) includes a plurality of exterior support cavities **710**, interior support cavities **720**, and/or other elements. The exterior support cavities **710** and interior support cavities **720** may be bounded by star-shaped posts **730** (e.g., star-shaped elements, cross-shaped contoured posts). The star-shaped posts **730** may include a diamond-shaped inner portion **740** (e.g., square and/or rectangular shaped inner portion) and star point elements **760**, **770** (e.g., star point elements) that extend away from the center of the post **730** at, for example, ninety degree angles (e.g., orthogonal to one another) to form the rough shape of a star and/or cross. In certain cases, star point elements **760** extending toward another post may be smaller in size than star point elements **770** extending towards an edge of the bottle packaging assembly. The larger star point elements **770** may extend further to encapsulate a portion of the outer surface of a bottle, thereby separating adjacent bottles and preventing adjacent bottles from contacting one another. In certain cases, a center support **700** includes recessed elements **780** between the posts **730**. The recessed elements **780** between posts **730** allow the posts **730** to tilt relative to one another and the center support **700** to flex during use. In certain cases, one or more posts **730** include a hole **750** to reduce a vacuum when multiple center supports are stacked (e.g., during manufacture).

(56) In some embodiments, a center support **700** is placed on a bottom tray (e.g., bottom tray **110** of FIGS. 1-3) when, for example, beverage containers are prepared for shipping. In some cases, the star-shape elements **730** are placed on posts included in a bottom tray (e.g., posts **114** of bottom tray **110** of FIG. 3). During assembly a flat surface of each post **730** contacts a flat surface of a post on the bottom tray (e.g., posts **114** or deformable post elements **116** of bottom tray **110** of FIG. 3).

The bottom tray **110** may include six posts to accommodate the six posts **730** of the center support **700**.

(57) FIG. **8** depicts a top tray of a beverage container packaging assembly according to some embodiments. In the example shown, a top tray **800** (e.g., top tray **120** of FIG. **1**) includes multiple vertical posts **810**. In certain cases, the vertical posts **810** form the sides (bounds) of bottle neck accommodating spaces **820** in the top tray **800**. The bottle neck accommodating spaces are configured to accommodate a neck of a beverage container, such as a wine bottle. The number of bottle neck accommodating spaces **820** corresponds to a number of beverage containers the top tray **800** is configured to accommodate. In the example shown, the top tray **800** includes twelve bottle neck accommodating spaces **820**—three (3) rows of four (4) bottle neck accommodating spaces—and the top tray **800** is configured to accommodate twelve beverage containers.

(58) FIG. **9** depicts a first view of a top tray according to some embodiments. In the example shown, a top tray **900** includes multiple vertical posts **910** that form bottle neck accommodating spaces **920**. For example, the vertical posts **910** may include deformable elements **912** that contact the neck of a bottle and hold it in place during shipping. The deformable elements **912** may also deform to absorb lateral loads applied to a bottle during transit. In some cases, the vertical posts **910** may include trapezoidal deformable elements **914** (e.g., protrusions) on a top of the vertical post **910**. The trapezoidal elements **914** may contact portions of the center support structure, such as corners **632** of the star-shaped cavities **630** of FIG. **6**.

(59) In certain cases, a top end of the bottle neck accommodating spaces **920** includes a circular depression **930** (e.g., circular depressed region). The circular depression **930** may accommodate the size of wine cork such that the wine bottle rim contacts the top tray but not the cork, for example, when a load is applied down on the top tray **900**. The circular depression **930** may include a thin deformable element bisecting the circular depression **930**.

(60) In some embodiments, the top tray **900** includes one or more holes **940**. Similar to the holes discussed with respect to other components of bottle packaging assembly, the holes **940** reduce vacuum generated when the top tray is placed into a box.

(61) FIG. **10** depicts a second view of a top tray according to various embodiments. In the example shown, a top tray **1000** includes the structure underlying the bottle neck accommodating spaces (e.g., bottle neck accommodating spaces **920** of FIG. **9**). The top side (on assembly) of the top tray **1000** includes non-contoured pulp paper. The top side of the bottle neck accommodating spaces may include cross-shaped protrusions **1010** (e.g., four protrusions meeting at a point in the example shown). The cross-shaped protrusions **1010** may absorb impact and reduce loads applied to the beverage containers during, for example, an impact event.

(62) FIG. **11** depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments. In the example shown, a beverage container **1140** is seated in a cup-shaped element of a bottom tray **1110**. A center support **1130** is seated upon the lower tray **1110**. The lower tray **1110** is similar to bottom tray **110** of FIGS. **1-3**, bottom tray **400** of FIG. **4**, and bottom tray **500** of FIG. **5**. One difference being the bottom tray **1010** is configured to accommodate fifteen beverage containers—three rows of five containers—as opposed to twelve in the bottom trays **110**, **400**, **500** of FIGS. **1-5**. The center support **1030** is similar to the center supports **130**, **600**, **700** of FIGS. **2**, **6**, and **7**, respectively, with the exception that center support **1030** is configured to accommodate fifteen beverage containers—three rows of five containers—as opposed to twelve.

(63) FIG. **12** depicts a bottom tray according to various embodiments. The bottom tray **1200** depicted is similar to bottom trays to bottom tray **110** of FIGS. **1-3**, bottom tray **400** of FIG. **4**, and bottom tray **500** of FIG. **5**. For the sake of brevity and clarity, the following description will focus primarily on the differences relative to the aforementioned bottom trays **110**, **400**, and **500**. In the example shown, a bottom tray **1200** includes a plurality of cup-shaped elements **1210**, posts **1230** (e.g., eight posts). The cup shaped element **1210** includes multiple thin deformable elements **1212**.

In certain cases, the thin deformable elements **1212** extend from a center of the cup-shaped element up a side of the cup-shaped element **1210**. The thin deformable elements terminate at the bottom of the cup-shaped element **1210** in a cross shape end **214** (e.g., cross shape element). The cross-shaped ends **1214** may assist in absorbing impact from a bottom of bottle when, for example, an assembly is dropped. The cross-shaped ends **1214** may reduce damage to the end of thin deformable element **1212** when, for example, the bottom tray **1200** is bent. The cross-shaped ends **1214** eliminate a stress concentration point and allow load to be distributed to other elements when the bottom tray **1200** is bent or otherwise loaded.

(64) In various embodiments, the cup-shaped elements **1210** elements include rounded corners **1220** at a base of the cup-shaped elements **1210**. The rounded corners **1220** may allow the bottom tray **1200** to endure more repeated loading. The rounded corners **1220** also accommodate the shape of the bottom of a bottle.

(65) FIG. 13 depicts a center support according to various embodiments. The center support **1300** shown is similar to the center supports **130**, **600**, **700** of FIGS. 2, 6, and 7, respectively, with the difference being the center support **1300** is configured to accommodate fifteen beverage containers—three rows of five containers—as opposed to twelve.

(66) FIG. 14 depicts a top tray according to various embodiments. The top tray **1400** shown is similar to top trays **120**, **800**, **900**, and **1000** of FIGS. 1, 8, 9, and 10. One difference between the top tray **1400** depicted is configured to accommodate fifteen beverage containers—three rows of five containers—as opposed to twelve as in top trays **120**, **800**, **900**, **1000**. In certain cases, top tray **1400** includes a plurality of posts **1410** each including holes **1420** in a top flat surface of the posts **1410**. The holes **1420** reduce any vacuum effect as the top tray is loaded into a box.

(67) FIG. 15 depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments. In the example shown, a beverage container **1540** is seated in a cup-shaped element of a lower tray **1510**. A center support **1530** is seated upon the bottom tray **1510**. The bottom tray **1510** is similar to bottom tray **1110**, **1200** of FIGS. 11 and 12. Center support **1530** is similar to center support **1300** of FIG. 13, with the exception that the center support **1530** includes bottle enclosure cavities **1532** that fully encapsulate each beverage container **1540**.

(68) In various embodiments, the beverage container enclosure cavities **1532** protect the beverage container **1540** from side impacts, maintain the beverage container **1540** upright during transit, and/or provide other benefits. In certain cases, a bottle packaging assembly (e.g., bottom tray **1510**, center support **1530**, and top tray **1400** of FIG. 14) may be able withstand 15 drops from at least 18 inches, two drops from 36 inches, and/or 10 drops from 30 inches, as required by various shipping companies (e.g., United Parcel Service (UPS), FedEx, and the like).

(69) FIG. 16 depicts a center support according to various embodiments. In the example shown, a center support **1600** includes a plurality of beverage container enclosure cavities **1610** (e.g., interior support cavities). The enclosure cavities **1610** function to maintain the bottle upright and to separate each bottle from adjacent bottles. The enclosure cavities **1610** may include a circular (cylindrical) enclosure. The enclosure cavities **1610** may fully encapsulate (surround) a neck, shoulder, body and/or other component of a beverage container. Walls **1620** of the enclosure cavities **1610** may include a draft angle, such as seven degrees and/or another draft angle.

(70) In some embodiments, the walls **1620** of the enclosure cavities **1610** include deformable elements **1622**. The deformable elements **1622** are configured to absorb load applied to the side of a beverage container, thereby protecting beverage container from damage during transit. Because the beverage containers are each fully encapsulated by enclosure cavities **1610**, the center support **1600** and beverage containers included therein may withstand higher loads and/or more load cycles than center support **1300** of FIG. 13.

(71) FIG. 17 depicts a bottom view of a center support according to various embodiments. In the example shown, a bottom side of center support **1700** (e.g., center support **1600** of FIG. 16, center support **1530** of FIG. 15) includes a plurality of beverage container enclosures **1710** (e.g., enclosure

cavities). The beverage container enclosure cavities **1710** may be bounded by star-shaped posts **1720** (e.g., star-shaped elements) and/or partial star-shaped posts **1730** (e.g., partial star-shaped elements). The star-shaped posts **1720** may include a diamond-shaped inner portion **1740** (e.g., square and/or rectangular shaped inner portion) and four outer portions **1750** (e.g., star point elements, cross arm elements) extend away from the center of the post **1720** at rough ninety degree angles (e.g., orthogonal to one another) to form the rough shape of a cross. In certain star point elements **1750** may include different sizes.

(72) In certain cases, a center support **1700** includes recessed elements **1760** elements between posts **1720**. The recessed elements **1760** between posts **1720** allow the posts **1720** to tilt relative to one another and the center support **700** to flex during use. In certain cases, one or more posts **1720** include a hole **1770** to reduce a vacuum when multiple center supports are stacked (e.g., during manufacture).

(73) In some embodiments, a center support **1700** is placed on a bottom tray element (e.g., bottom tray element **1110**, **1200**, **1510** of FIGS. **11**, **12**, and **15**) when a beverage container packaging assembly is prepared for shipping. In some cases, the posts **1720** are placed on posts included in a bottom tray (e.g., posts **114** and/or platforms **116** of bottom tray **110** of FIG. **3**). During assembly a flat surface of each post **1720** contacts a flat surface of a post on the bottom tray (e.g., posts **1230** of bottom tray **1200** of FIG. **12**). The bottom tray **1200** may include eight posts **1230** to accommodate the eight internal posts **1720** of the center support **1700**.

(74) FIG. **18** depicts a beverage container packaging assembly according to various embodiments. In the example shown, a beverage container packaging assembly **1800** may include a bottom tray **1810** (e.g., bottom element), a top tray **1820** (e.g., a top element), a center support **1830** (e.g., partition element, center support element), and/or other components. The beverage container packaging assembly **1800** is configured to package a plurality of beverage containers **1840** for transit. The base of the beverage container **1840** sits in one of the cylindrical cup-shaped elements of the bottom tray **1810**. The center support **1830** separates the bottom tray **1810** and top tray **1820**. The top tray **1820** rests on the center support **1830**. The top of the beverage container **1840** is encapsulated in a rectangular cup-shaped element of the top tray **1820**. The center support **1830** prevents the sides of the beverage containers **1840** from contacting during shipping and provides spacing between the bottom tray **1810** and top tray **1820**. In certain cases, the center support **1830** may not be included in the beverage container packaging assembly **1800**.

(75) FIG. **19** depicts a bottom tray and center support of a beverage container packaging assembly according to various embodiments. In the example shown, a beverage container **1940** (e.g., wine bottle) is seated in a cup-shaped element **1912** of a lower tray **1910**. The cup-shaped element **1912** is shaped to retain the beverage container **1940** in a vertical configuration during transit. In the example shown, a center support **1930** includes multiple cardboard elements—two laterally arranged cardboard elements **1932**, **1934** and two transversely arranged cardboard elements **1936**, **1938**.

(76) FIG. **20** depicts a bottom tray of a beverage container packaging assembly according to some embodiments. As shown, a bottom tray **2010** includes a plurality of cup-shaped elements **2012**. The number of cup-shaped elements **2012** corresponds to a number of beverage containers the bottom tray **2010** is configured to accommodate. In the example shown, the bottom tray **2010** includes twelve cup-shaped elements **2012**—three (3) rows of four (4) cup-shaped elements **2012**—and the bottom tray **2010** is configured to accommodate 12 beverage containers. FIG. **20** depicts one example configuration of cup-shaped elements. The present disclosure, however, is in no way limited to the depicted configuration of cup-shaped elements **2012** or number of cup-shaped elements **2012**.

(77) In certain cases, the bottom tray **2010** includes posts **2014** configured to support the center support (not shown). Each of the posts **2014** may include guide elements **2016** (e.g., protrusions) that form one or more channels to accommodate panels of the center support (not shown).

(78) FIG. 21 depicts cup-shaped elements of a bottom tray according to various embodiments. In the example shown, a bottom tray **2100** (e.g., bottom tray **2010** of FIG. 20) may include cup-shaped elements **2110** (e.g., cup shaped element **1912**, **2012** of FIGS. 19, 20) that are cylindrical and/or substantially cylindrical. The cup-shaped element may include a bottom portion and multiple side walls forming a cup shape. In certain cases, the side walls may not be exactly vertical but may include draft to enable the bottom tray **2100** to be easily removed from a mold during fabrication.

(79) Each of multiple cup-shaped elements **2110** includes a first deformable element **2112** (e.g., an hourglass shaped element, bow-tie shaped element), second deformable elements **2116**, vertical elements **2118**, **2120**, **2122**, **2124**, and other elements. The first deformable element **2112** may be raised (e.g., a protrusion) from the bottom surface of the cup-shaped element **2110**. The first deformable element **2112** may resemble an hourglass, bow tie, or other similar shape. The first deformable element **2112** is configured to absorb impact applied to a top of a beverage container (not shown). The first deformable element **2112** may be particularly effective in absorbing impact from a bottle with a convex bottom surface, such as a wine bottle. Wider portions **2114** of the hourglass shaped element **2112** are configured to contact the outer edges of the bottom of a bottle, such as a wine bottle. Thinner portions **2113** of the hourglass shaped element **2112** allow the element to deform (for example, by bending, crumpling, and/or otherwise deforming) when a load is applied to a beverage container housed in the bottom tray **2100**. The hourglass shaped element **2112** deforms to absorb load and reduce damage to a bottle, particularly when a load is applied down from the top on the bottle or when the assembly is dropped.

(80) A cup-shaped element **2110** may include second deformable elements **2116**. The second deformable elements **2116** may resemble two adjacent mountain peaks. Similar to the first deformable element **2112**, the second deformable elements **2116** are configured to contact the outer edges of the bottom of a bottle, such as a wine bottle. When a load is applied to the top of the wine bottle, when the beverage container packaging is dropped, or when the bottle is otherwise subjected to a force, the second deformable elements **2116** are configured to absorb the load and/or energy of the load by, for example, crushing, buckling, and/or otherwise deforming.

(81) In various embodiments, the cup-shaped element **2110** includes multiple vertical deformable elements **2118**, **2120**, **2122**. The vertical deformable elements **2118**, **2120**, **2122** contact the sides of a beverage container (not shown). In certain cases, first vertical elements **2118**, second vertical elements **2120**, and third vertical elements **2122** are configured to contact the outside of a bottle. The first vertical elements **2118**, second vertical elements **2120**, and third vertical elements **2122** may be sized, such that a bottle contacts all of the vertical elements **2118**, **2120**, **2122** when loaded into the cup-shaped element **2110**. The bottle (not shown) and vertical deformable elements **2118**, **2120**, **2122** may contact one another in an interference fit, such that a force is necessary to push the bottle into the cup-shaped element **2110** and a force is necessary to remove the bottle from the cup-shaped element **2110**. Securing the bottle in the cup-shaped element **2110** in such a manner ensures that the bottle is stationary during transit, thereby reducing any potential damage.

(82) According to some embodiments, the cup-shaped element **2110** includes multiple thin deformable elements **2124**, **2126**. In certain cases, four thin deformable elements **2124** may extend from a center of the cup-shaped element up a side of the cup-shaped element **2110**. The four first deformable elements **2124** may include thin protrusions spaced roughly 90 degrees apart from one another. The thin deformable elements **2124** may collectively form an X-shape, with each thin deformable element **2124** extending from the center of the X along the bottom and up a side wall. In certain cases, a portion of a first deformable element **2124** extending along the wall of the cup-shaped element **2110** may extend further from the surface than a portion of the thin deformable element spanning the bottom of the cup-shaped element **2110**. In other words, the portion of the thin deformable element **2124** spanning the wall may be taller (higher) than the portion spanning the bottom of the cup-shaped element **2110**. In some embodiments, second thin deformable

elements **2126** may be disposed between vertical elements **2118**. Similar to the vertical elements, thin deformable elements **2124**, **2126** are configured to absorb impact energy and/or loads applied to the sides of the bottle. The thin deformable elements **2124**, **2126** prevent damage to the bottom and sides of the bottle.

(83) FIG. **22** depicts a bottom tray of a beverage container packaging assembly according to various embodiments. A first cross-section A-A depicts a cross-section of the cup-shaped element **2212** of the bottom tray in a first direction. The first cross-section A-A is depicted in FIG. **23**. A second cross section B-B depicts a cross-section of the cup-shaped element **2212** in a direction perpendicular to the first direction. The second cross-section B-B is depicted in FIG. **24**.

(84) FIG. **23** depicts a first cross-section view of a cup-shaped element of a bottom tray according to various embodiments. FIG. **23** includes a cross-section view along section A-A as shown in FIG. **22**. In the example shown, the cross-section passes through the center of the hourglass shaped deformable element **2312**. The center of the hourglass shaped deformable element **2312** includes a thinner portion **2313** of the hourglass shaped deformable element **2312**. The wider portion **2314** of the hourglass shaped deformable element **2312** is shown in the background. The wider portion **2314** contacts the outer edges of the bottom of a beverage container. The second deformable elements **2316** also contact the outer edges of the bottom of the beverage container (not shown). First vertical elements **2318** contact the sides of the beverage container to retain the container in place. Thin deformable elements **2324** extend from the center of the cup-shaped element up the sides of the cup-shaped element. In certain cases, a thin deformable element **2326** may be disposed between the first vertical elements **2316**.

(85) FIG. **24** depicts a second cross-section view of a cup-shaped element of a bottom tray according to various embodiments. FIG. **24** includes a cross-section view along section B-B as shown in FIG. **22**. In the example shown, the cross-section passes through the center of the hourglass shaped deformable element **2412**. The thinner portions **2413** and wider portions **2414** of the hourglass shaped element **2412** are raised up from a bottom surface of the cup-shaped element **2412**. The wider portions of the hourglass shaped element **2412** contact a beverage container placed into the cup-shaped element **2412**. When a load is applied to a beverage container with a convex bottom (such as a wine bottle, beer bottle, etc.), the wider portions **2414**, which contact the bottom of the bottle, are configured to deform and absorb the energy of the load. The second deformable portion **2416** (shown in the background of the cross-section) is similarly configured to deform and absorb a load applied to a beverage container.

(86) FIG. **25** depicts a top tray of a beverage container packaging assembly according to some embodiments. In the example shown, a top tray **2520** includes rectangular cup-shaped elements **2522**, vertical posts **2524**, and/or other elements. The rectangular cup-shaped elements **2522** are configured to accommodate a neck of a beverage container, such as a wine bottle. The number of cup-shaped elements **2522** corresponds to a number of beverage containers the top tray **2520** is configured to accommodate. In the example shown, the top tray **2520** includes twelve rectangular cup-shaped elements **2522**—three (3) rows of four (4) cup-shaped elements **2522**—and the bottom tray is configured to accommodate twelve beverage containers. FIG. **25** depicts one example configuration of rectangular cup-shaped elements, and the present disclosure is in no way limited to the depicted configuration.

(87) FIG. **26** depicts a top tray of a beverage container packaging assembly according to some embodiments. In the example shown, a top tray **2620** includes rectangular cup-shaped elements **2622**, vertical posts **2624**, and/or other elements. The rectangular cup-shaped elements **2622** include four walls forming roughly the shape of a rectangular enclosure. The vertical posts **2624** include vertical deformable elements **2626** (on each of the four surrounding vertical posts **2624**). A neck of bottle may, for example, be in contact with four vertical deformable elements **2626**. The vertical deformable elements **2626** in contact with the neck of the bottle restrain the bottle from movement during shipment. The vertical deformable elements **2626** may also absorb loads applied

to a side of the bottle.

(88) In various embodiments, a bottom portion of the rectangular cup-shaped element **2622** includes a cylindrical depression **2628**. The cylindrical depression **2628** may be sized to accommodate a top of a bottle (not shown) and to restrain the top of the bottle from movement during transit.

(89) FIG. **27** depicts a bottom tray and center support of a beverage container packaging assembly according to some embodiments. In the example shown, a center support **2730** rests on a lower tray **2710** (e.g., lower tray **1810** of FIG. **18**). The center support **2730** may be an alternative design relative to center support **1830**, **1930** of FIGS. **18** and **19**. The center support **2730** includes two cardboard elements **2732**, **2734**. The cardboard elements **2732**, **2734** may include corrugated cardboard or any other type of cardboard. The cardboard elements **2732**, **2734** may be L-shaped and/or include a bend. A first cardboard element **2732** may include cutouts (e.g., notches) that align with cutouts in the second cardboard element **2734**. When installed, the first cardboard element **2732** and second cardboard element **2734** may interlock to form a rectangular section **2736**. The rectangular section **2736** may span (e.g., surround, enclose) two securement chambers of the lower tray **1810**.

(90) FIG. **28** depicts a beverage container packaging assembly including a lower tray and center support according to some embodiments. In the example shown, a center support **2830** rests on a lower tray **2810** (e.g., lower tray **1810** of FIG. **18**). The center support **2830** may be an alternative design relative to center support **1830**, **1930** of FIGS. **18** and **19** and the center support **2730** of FIG. **27**. The center support **2830** includes two cardboard elements **2832**, **2834**. Each of the cardboard elements **2832**, **2834** includes two corrugated sections. In other words, the cardboard elements **2832**, **2834** include double-walled corrugated cardboard including two layers of corrugation. For example, the two layers of corrugation may resemble double-layered sandwich.

(91) In various embodiments, the center support **2830**, the center support **2730** of FIG. **27**, center support **1830** of FIG. **18**, center support **600** of FIG. **6**, center support **700** of FIG. **7**, center support **1300** of FIG. **13**, center support **1600** of FIG. **16**, and center support **1700** of FIG. **17** include several example center support configurations contemplated by the present disclosure. In some embodiments (not shown), the center support may include triple-walled corrugated and/or any other number of stacked corrugated sections. Alternatively, the center support may include cardboard without any corrugation. For example, the center support may include a stack of multiple sheets of cardboard with no corrugation. The present disclosure, however, is not limited to these configurations and is intended to encompass a wide variety of center support designs.

(92) In various embodiments, assembly of a beverage container packaging assembly **2800** is depicted. A lower tray **2810** may be placed into a container **2850**, such as a cardboard box, crate, and/or other container. A center support (e.g., center support **600**, **700**, **1300**, **1600**, **1700**, **1830**, **2730**, **2830**, and/or any other center support) is placed onto the lower tray **2810**. Beverage containers (not shown) are loaded into cup-shaped elements of the lower tray **2810**. An upper tray (not shown) is placed on the top of the center support. The container **2850** is sealed by, for example, closing the flaps of the box and/or applying tape. The container **2850** may be then be shipped to its recipient. And upon receipt, a recipient may perform the inverse (opposite) of these steps to unpack the beverage container packaging assembly **2800**.

(93) In various embodiments, these and other steps to assemble and disassemble a container packaging assembly **2800** may be performed in other sequences to achieve similar results.

(94) While various embodiments of the present invention have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of the present invention should not be limited by any of the above-described illustrative embodiments, but should instead be defined only in accordance with the following claims and their equivalents.

Claims

1. A beverage container packaging assembly comprising: a bottom tray of molded paper pulp including a plurality of cup-shaped elements, the cup-shaped elements including a first deformable element and a second deformable element, the first and second deformable elements including different widths, wherein at least one of the first deformable element and the second deformable element includes at least two overlapping elements; a top tray of molded paper pulp including a plurality of bottle neck accommodating spaces each bounded by at least one deformable element, the bottle neck accommodating spaces configured to accommodate a neck portion of the beverage container; and a center support of molded paper pulp disposed between the bottom tray and the top tray, the center support including a plurality of center support posts, at least one recessed element located between at least two of the center support posts, and a plurality of beverage container support cavities bounded by one or more of the center support posts, wherein one or more walls of the beverage container support cavities are configured to surround at least a portion of the beverage container, and wherein a cross-sectional area of the center support decreases between a top end of the center support and a bottom end of the center support.
2. The beverage container packaging assembly of claim 1, wherein the at least two overlapping elements form a cross shape.
3. The beverage container packaging assembly of claim 1, wherein the first deformable element and the second deformable element include a cross shape.
4. The beverage container packaging assembly of claim 1, wherein the top tray further includes a plurality of vertical members, and the center support posts contact one of the plurality of vertical members when the beverage container packaging assembly is assembled.
5. The beverage container packaging assembly of claim 4, wherein at least one first flat surface of the center support posts contacts at least one second flat surface of the vertical members.
6. The beverage container packaging assembly of claim 1, wherein the cup-shaped elements comprise a draft.
7. The beverage container packaging assembly of claim 1, wherein the cup-shaped elements vary in diameter from a bottom end to a top end of the bottom tray.
8. The beverage container packaging assembly of claim 1, wherein the top tray further includes a plurality of vertical members including one or more deformable elements, the deformable elements at least partially surrounding the bottle neck accommodating spaces.
9. The beverage container packaging assembly of claim 8, wherein at least one of the vertical members comprises a vertical post.
10. The beverage container packaging assembly of claim 8, wherein the deformable elements extend towards a center of at least one of the bottle neck accommodating spaces.
11. The beverage container packaging assembly of claim 1, wherein the beverage container support cavities include one or more of a quarter circle shape, a half circle shape, and full circle shape.
12. The beverage container packaging assembly of claim 1, wherein at least one of the beverage container support cavities include a semi-circular shape and at least one of the beverage container support cavities include a fully circular shape.
13. The beverage container packaging assembly of claim 1, wherein the walls of the beverage container support cavities are configured to fully encapsulate the beverage container.
14. The beverage container packaging assembly of claim 1, wherein a first set of the beverage container support cavities are configured to fully encapsulate the beverage container and a second set of the beverage container support cavities are configured to partially encapsulate the beverage container.
15. The beverage container packaging assembly of claim 1, wherein the bottom tray, center support, and top tray are configured to contact the beverage container.

16. The beverage container packaging assembly of claim 1, wherein at least one of the center support posts include a thru hole.

17. The beverage container packaging assembly of claim 1, wherein the beverage container support cavities are configured to fully encapsulate a beverage container.

18. The beverage packaging assembly of claim 1, wherein the beverage support cavities further include a draft such that the beverage support cavities vary in diameter from a bottom end of the center support to a top end of the center support.
